

Franco David Barrionuevo

Master of Spatial Information Applications (MAIE) Student · Córdoba, Argentina

Personal and Contact Information

Date of Birth: 11/04/1995

Birth Place: Buenos Aires, Argentina

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Education

Master of Spatial Information Applications (MAIE)

Instituto Gulich. Córdoba, Argentina

02/2026 – In progress

Courses completed and approved: Mathematics, Introduction to Remote Sensing, Introduction to intelligent methods for planning, scheduling, and execution problem solving, Statistics, and Numerical Methods Applied to Satellite Data Processing.

Remaining Coursework: Environmental Remote Sensing, Digital Processing of Satellite Imagery and Geographic Information Systems (GIS), Applications of Synthetic Aperture Radar (SAR) Imagery, Numerical Models for Early Warning Systems, Risk Mapping, and Simulation, Spatial Analysis and Risk Assessment, Environmental Assessment, Monitoring, and Management Tools, Research Methodology and Thesis Development Tools, and Integrative Project.

Environmental Engineering

Universidad Nacional de San Martín (UNSAM). Buenos Aires, Argentina

2017 – 2025

Courses and topics covered Specialization: Sanitary and Waste Treatment Engineering.

Optional courses: Programming, Data Science and Machine Learning, GIS, and Remote sensing.

Final Project Title: Prefeasibility Study for the Installation of an Anaerobic Biodigester for the Codigestion of Pig Manure and Nopal (*Opuntia ficus-indica*) in Rosario Vera Peñaloza, La Rioja.

Medicine

Universidad de Buenos Aires (UBA). Buenos Aires, Argentina

2017 – 2025

Complementary Courses

Diploma in Data Science and Artificial Intelligence

Universidad Nacional de San Martín (UNSAM). Buenos Aires, Argentina

02/2023 – 10/2023

Artificial Intelligence and Quantum Computing for Earth Observation

Gulich Institute, Geoscience and Remote Sensing Society (GRSS) & Pavia University. Córdoba, Argentina

10/2025

Introduction to High Performance Computing in Python

Gulich Institute. Córdoba, Argentina

09/2025

Remote Sensing as a Tool for Earth and Forest Resource Monitoring: Uses and Applications of the SAR SAOCOM and Sentinel-1 Missions

Gulich Institute, GRSS & European Space Agency (ESA). Córdoba, Argentina

10/2024

Computational Tools for Climate Science

Climatematch Academy. Remote

07/2024

Work Experience

CONAE Fellow

Instituto Gulich

02/2026 – Present

Full-time, fully funded scholarship recipient to attend the Master of Spatial Information Applications (MAIE) program courses at the Instituto Gulich, located at the Teófilo Tabanera Space Center (CETT).

Research topic: Implementation of an Air Quality Model System, and its Evaluation Using Satellite Data.

Climate Change Consultant

Escuela de Hábitat y Sostenibilidad (EHyS-UNSAM)

07/2025 – 01/2026

- Reconstructed the Marine Heatwaves (MHW) historical record from 1980 to 2020 in the Brazilian Current to assess extreme climate events and ecological impacts in the Buenos Aires Coast area.
- Implemented large geospatial data processing workflows in Python for a wide variety of ocean data sources: bathymetry, SST interpolated data, ocean reanalysis and atmospheric reanalysis.

Anaerobic Digestion Technologist

Instituto Nacional de Tecnología Industrial (INTI)

06/2023 – 02/2026

- Led projects on organic wastes mapping for biodigestion generation using data science and GIS tools.
- Developed tailored software in Python to automatize biological laboratory test data processing.
- Performed Biochemical Methane Potential (BMP) tests on conventional and non-conventional organic substrates (e.g., microalgae, nopal).

Undergraduate Research Fellow

Universidad Nacional de San Martín (UNSAM)

02/2024 – 08/2025

- Implemented geoprocessing tools in Python, Google Earth Engine, and GIS software to analyze and visualize satellite-derived data from various sources.
- Reconstructed historical paddy fields expansion implementing a timeseries clustering, evaluated historical human-caused fires, and assessed air quality affection caused by fires through satellite-derived tropospheric air pollutants measurements.

Impact Scholar

Climatematch Academy

10/2023 – 03/2024

- Implemented multi-dimensional arrays handling tools in Python for large geospatial data processing.
- Programmed algorithms in Python for processing satellite-derived burned area products and climate reanalysis data.
- Authored a scientific publication detailing the methodology and results obtained during the project's duration.

Undergraduate Research Fellow

Instituto Nacional de Tecnología Industrial (INTI)

08/2022 – 07/2023

- Conducted research on the national availability of organic wastes from industrial processes for utilization in biogas production. Mapped the national distribution of organic waste-producing industries in Argentina.
- Performed laboratory work, including the characterization of anaerobic sludges and organic substrates, and conducted Biochemical Methane Potential (BMP) tests.

Publications

Barrionuevo, F., Rocha, R., Rojas, L., Rengifo, R., & Farfán C. (2024). *Latitudinal Assessment of Burned Area within the Argentinian Andean-Patagonian Forest for the 2001-2019 Fire Seasons*. <https://doi.org/10.5281/zenodo.12206635>

Conferences and Seminars

Oral Presentations

Geomatics in Environmental Monitoring GEM-24 Workshop

Virtual | 2024

Title: *Uso de técnicas de clustering sobre series temporales de índices verdes MODIS para la delimitación de complejos arroceros en el Bajo de los Saladillos (Santa Fe).*

Climatematch Impact Scholars Program 2023-2024 Seminars

Virtual | 2024

Title: *Latitudinal Assessment of Burned Area within the Argentinian Andean-Patagonian Forest for the 2001-2019 Fire Seasons.*

Poster Presentations

V Simposio de Residuos

Mar del Plata, Argentina | 2025

Title: *A Methodology for Mapping and Use in Biogas Production.* **Barrionuevo, Franco;** Videla, Estefanía; Farías, Camila; Flesler, Fanny; Rodríguez, Ruth.

Tropentag 2025

Bonn, Germany | 2025

Title: *Land use changes and anthropogenic impacts due to rice fields in Santa Fe (Argentina).* Clauser, C. D., Peluso, J., Aronzon, C., **Barrionuevo, F.**, Ali Santoro, V., Morandeira, N.

VII Congreso Nacional de Conservación de la Biodiversidad

San Juan, Argentina | 2025

Title: *Degradación de las vegas y bosquecitos de chacay por cambios de cobertura e incendios en la cuenca del río Blanco (Mendoza, Argentina).* Torre Azzaroni, María Luisina; **Barrionuevo, Franco David;** Rubio, María Cecilia; Morandeira, Natalia Soledad.

Expo EHyS 2024 (EHyS-UNSAM)

Buenos Aires, Argentina | 2024

Title: *Enfoque satelital para evaluar quemas en arrozceras.* **Barrionuevo, Franco David;** Clauser, Carlos David; Morandeira, Natalia Soledad.

IV Simposio de Residuos

Mendoza, Argentina | 2023

Title: *Participación en programa internacional de ensayos de aptitud de potencial metanogénico de sustratos.* Videla Estefanía; Camila Farias; Diego Rodriguez; Fanny Flesler; **Franco Barrionuevo;** Ruth Rodriguez.

Title: *Valorización de residuos orgánicos y subproductos industriales a partir del potencial bioquímico metanogénico.* Ruth Rodriguez; Estefanía Videla; Camila Farias; Diego Rodriguez; Fanny Flesler; **Franco Barrionuevo.**

Language Skills

Spanish (Native)

English (Reading: C1, Writing: C1, Speaking: B2)

Portuguese (Reading: A2, Writing: A2, Speaking: A2)

Technical Skills and Competences

Programming Languages: Python, Fortran, R and JavaScript.

Remote Sensing and Geospatial Technologies: QGIS, Google Earth Engine (GEE), STAC API, Satellite data processing and analysis, and Time-series analysis of Earth Observation data.

Machine Learning and Data Science: Machine Learning and Deep Learning for Earth Observation, Convolutional Neural Networks (CNNs), U-Net architectures for semantic segmentation, and Environmental and geospatial data analysis.

Scientific Computing and Modeling: High-performance computing (HPC), Numerical modeling and data processing workflows, and Atmospheric and environmental modeling.

Software Development and Collaboration: Git/GitHub, Web mapping application development and Project management.